

Fig. 1: Transition region of the electromagnetic spectrum lying between the microwave and infrared regimes is the THz regime

imaging of skin cancer.

In the THz range, essentially two groups of near-field methods have been used till recently: one making use of sub-wavelength apertures and the other using sharp tips that scatter the THz radiation. In both cases, the resolution is directly linked to the size of the imaging tool. However, only a small fraction of the incident radiation contributes to the THz image.

One way to overcome the diffraction limit is to limit the detection area with an aperture. By using a detector with an integrated aperture, near-field images can be made with a spatial resolution determined by the aperture size and not the THz wavelength.

Generally, the near-field is an environment where the distance to an object of interest is 10-100 times larger than the physical size of the THz imaging array. Researchers have developed an aperture-less technique in which not the radiation scattered into the far field, but the near field close to the tip is measured. The advantage of near-field detection over far-field detection is that one can directly measure the enhanced field around the tip, resulting in relatively strong signals.

The method to obtain a sub-wavelength resolution has the following advantages:

1. The resolution is ultimately determined by the dimensions of the tip, not the incident terahertz beam size.
2. The measurements are background free.

3. The detection setup measures in the near field, where the electric field can be strong. This provides sufficiently large signals and thus reduces the measurement time.

4. The method has the more practical requirement of scanning a tip along a surface, instead of two surfaces paral-

lel to each other.

In general, the THz radiation can be used to create images or to communicate information, just as visible light can create a photograph, audible radio wave can transmit music and speech, or invisible radiation can reveal broken bones.

THz imaging can be elucidated using the concept of 2D spectroscopy. The principle is based on scanning the sample through the focus of a THz-time domain spectroscopy (THz-TDS) transmission or reflection spectrometer as shown in Fig. 3.

Ordinary spectroscopy measures the intensity of light at specific frequencies, whereas THz-TDS directly measures the temporal electric field. Fourier transformation of this time-domain data gives the amplitude and

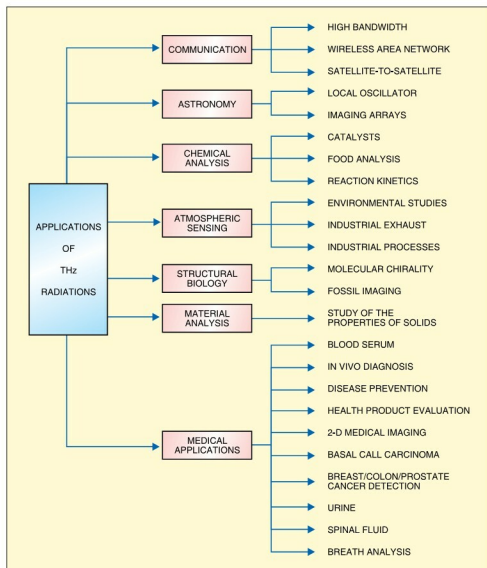


Fig. 2: Potential application areas of the THz electromagnetic radiations